

2016 HSC Mathematics Marking Guidelines

IMPORTANT:

This resource was developed to support a previous version of the syllabus.
It may contain content that differs from the current syllabus.

Section I

Multiple-choice Answer Key

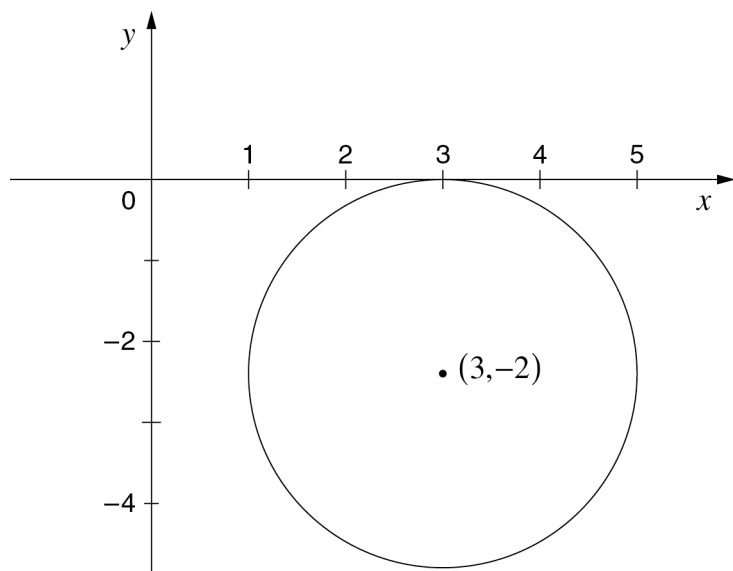
Question	Answer
1	B
2	C
3	B
4	A
5	B
6	A
7	A
8	D
9	C
10	D

Section II

Question 11 (a)

Criteria	Marks
• Provides correct sketch	2
• Identifies radius, or equivalent merit	1

Sample answer:



Question 11 (b)

Criteria	Marks
• Provides correct derivative	2
• Attempts to use quotient rule, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \frac{d}{dx} \left(\frac{x+2}{3x-4} \right) &= \frac{(3x-4)(1) - (x+2)(3)}{(3x-4)^2} \\
 &= \frac{3x-4-3x-6}{(3x-4)^2} \\
 &= \frac{-10}{(3x-4)^2}
 \end{aligned}$$

Question 11 (c)

Criteria	Marks
• Provides correct solution	2
• Establishes that $x \leq 5$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 |x - 2| &\leq 3 \\
 \therefore -3 &\leq x - 2 \leq 3 \\
 -1 &\leq x \leq 5
 \end{aligned}$$

Question 11 (d)

Criteria	Marks
• Provides correct solution	2
• Correct primitive, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \int_0^1 (2x + 1)^3 dx &= \left[\frac{1}{2} \times \frac{1}{4} (2x + 1)^4 \right]_0^1 \\
 &= \frac{1}{8} (3)^4 - \frac{1}{8} (1)^4 \\
 &= \frac{81}{8} - \frac{1}{8} \\
 &= \frac{80}{8} \\
 &= 10
 \end{aligned}$$

Question 11 (e)

Criteria	Marks
• Provides correct solution	3
• Obtains $x^2 - 2x - 8 = 0$ and solves for x , or equivalent merit	2
• Attempts to eliminate x or y , or equivalent merit	1

Sample answer:

$$\left. \begin{array}{l} y = -5 - 4x \\ y = 3 - 2x - x^2 \end{array} \right\} \begin{array}{l} \textcircled{1} \\ \textcircled{2} \end{array}$$

Subs $\textcircled{1}$ into $\textcircled{2}$

$$-5 - 4x = 3 - 2x - x^2$$

$$x^2 - 2x - 8 = 0$$

$$(x - 4)(x + 2) = 0$$

$$\therefore x = 4 \text{ or } -2$$

$$\text{Subst } x = 4 \text{ into } \textcircled{1} \Rightarrow y = -5 - 4(4) = -21$$

$$\text{Subst } x = -2 \text{ into } \textcircled{1} \Rightarrow y = -5 - 4(-2) = 3$$

 $\therefore$ points of intersection are $(4, -21)$ and $(-2, 3)$.**Question 11 (f)**

Criteria	Marks
• Provides correct solution	2
• Obtains $\sec^2 \frac{\pi}{8}$, or equivalent merit	1

Sample answer:

$$y = \tan x$$

$$\frac{dy}{dx} = \sec^2 x$$

$$\text{When } x = \frac{\pi}{8}, \frac{dy}{dx} = \sec^2 \left(\frac{\pi}{8} \right)$$

$$= \frac{1}{\cos^2 \frac{\pi}{8}}$$

$$\approx 1.17$$

 $\therefore$ Gradient of tangent is 1.17.

Question 11 (g)

Criteria	Marks
• Provides correct solution	2
• Obtains one correct answer, or equivalent merit	1

Sample answer:

$$\sin\left(\frac{x}{2}\right) = \frac{1}{2} \quad \text{for} \quad 0 \leq x \leq 2\pi$$

$$\text{Note} \quad 0 \leq \frac{x}{2} \leq \pi$$

$$\therefore \left(\frac{x}{2}\right) = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$x = \frac{\pi}{3}, \frac{5\pi}{3}$$

Question 12 (a) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains correct slope, or equivalent merit	1

Sample answer:

$$\frac{y-4}{x-2} = \frac{1-4}{6-2}$$

$$\frac{y-4}{x-2} = \frac{-3}{4}$$

$$4(y-4) = -3(x-2)$$

$$4y - 16 = -3x + 6$$

$$3x + 4y - 22 = 0$$

Question 12 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to calculate perpendicular distance of a point from a line, or equivalent merit	1

Sample answer:

$$BC : 3x + 4y - 22 = 0$$

$$\begin{aligned}\therefore AD &= \frac{|3(1) + 4(0) - 22|}{\sqrt{3^2 + 4^2}} \\ &= \frac{|-19|}{\sqrt{25}} \\ &= \frac{19}{5}\end{aligned}$$

Question 12 (a) (iii)

Criteria	Marks
• Provides correct solution	2
• Finds distance from B to C , or equivalent merit	1

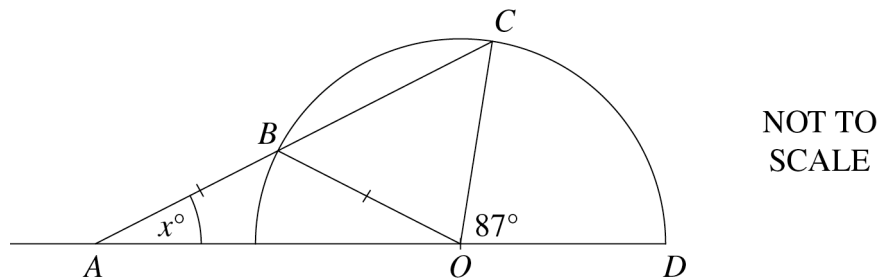
Sample answer:

$$\begin{aligned}BC &= \sqrt{(2-6)^2 + (4-1)^2} \\ &= \sqrt{16+9} \\ &= \sqrt{25} \\ &= 5\end{aligned}$$

$$\begin{aligned}\therefore \text{Area } \triangle ABC &= \frac{1}{2} \times 5 \times \frac{19}{5} \\ &= \frac{19}{2} \text{ square units}\end{aligned}$$

Question 12 (b) (i)

Criteria	Marks
• Provides correct explanation	1

Sample answer:

$$\angle BOA = x^\circ \quad (\angle\text{s opposite equal sides in } \triangle ABO)$$

$$\therefore \angle CBO = 2x^\circ \quad (\text{exterior } \angle \text{ of } \triangle ABO \text{ equal to sum of two opposite interior } \angle\text{s})$$

Question 12 (b) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to use the fact that triangle OBC is isosceles, or equivalent merit	1

Sample answer:

$$OC = OB \text{ (radii)}$$

$$\angle BCO = \angle CBO \text{ (}\angle\text{s opposite equal sides in } \triangle CBO\text{)}$$

$$= 2x^\circ$$

Then $\angle COD = \angle CAO + \angle BCO$ (exterior $\angle$ of $\triangle CAO$ equals sum of two opposite interior $\angle$ s)

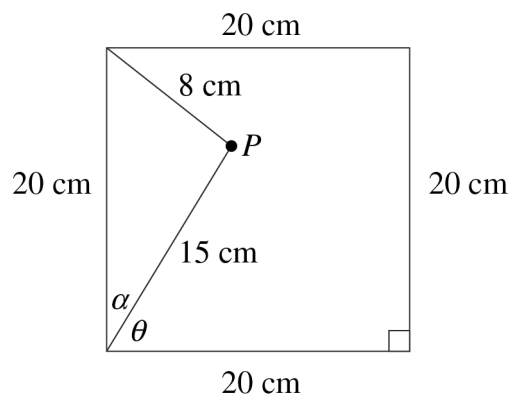
$$\therefore 87 = x + 2x$$

$$87 = 3x$$

$$x = 29$$

Question 12 (c)

Criteria	Marks
• Provides correct solution	3
• Substitutes correctly into cosine rule, and finds angle or equivalent merit	2
• Identifies the complementary angle to θ , or equivalent merit	1

Sample answer:

$$\begin{aligned}\cos \alpha &= \frac{15^2 + 20^2 - 8^2}{2 \times 20 \times 15} \\ &= \frac{561}{600} \\ \alpha &\approx 20.77^\circ \\ \theta &\approx 69^\circ \text{ to nearest degree.}\end{aligned}$$

Question 12 (d) (i)

Criteria	Marks
• Provides correct derivative	1

Sample answer:

$$y = xe^{3x}$$

$$\begin{aligned}\frac{dy}{dx} &= 3xe^{3x} + e^{3x} \\ &= e^{3x}(1 + 3x)\end{aligned}$$

Question 12 (d) (ii)

Criteria	Marks
• Provides correct solution	2
• Attempts to use part (i) or equivalent merit	1

Sample answer:

Find exact value of $\int_0^2 e^{3x}(3+9x)dx$

$$\begin{aligned}
 \int_0^2 e^{3x}(3+9x)dx &= \int_0^2 3e^{3x}(1+3x)dx \\
 &= 3 \int_0^2 e^{3x}(1+3x)dx \\
 &= 3 \left[xe^{3x} \right]_0^2 \\
 &= 3 \left[(2e^6) - (0e^0) \right] \\
 &= 6e^6
 \end{aligned}$$

Question 13 (a) (i)

Criteria	Marks
• Provides correct solution	4
• Finds the x -values at which the stationary points occur and verifies the maximum turning point, or equivalent merit	3
• Finds the x -values at which the stationary points occur, or equivalent merit	2
• Attempts to solve $\frac{dy}{dx} = 0$, or equivalent merit	1

Sample answer:

$$y = 4x^3 - x^4$$

Find stationary points and determine nature.

$$\frac{dy}{dx} = 12x^2 - 4x^3$$

$$\begin{aligned} \text{Need } \frac{dy}{dx} &= 0 & 12x^2 - 4x^3 &= 0 \\ & & 4x^2(3 - x) &= 0 \\ & & \text{when } x = 0 \text{ or } x = 3. & \end{aligned}$$

$$\begin{aligned} \text{when } x = 0 & \quad y = 0 \\ \text{when } x = 3 & \quad y = 27 \end{aligned}$$

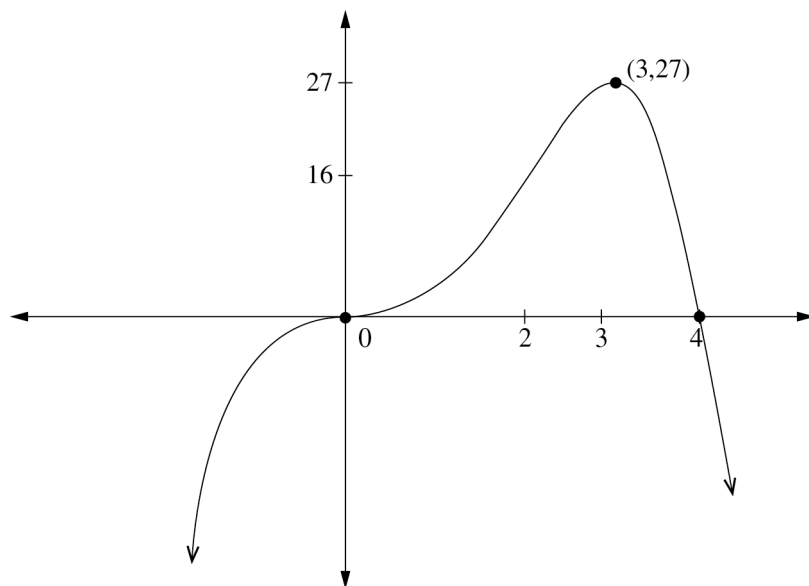
Checking the gradients for

$$x = 0, \quad \begin{array}{c|c|c|c} x & 0^- & 0 & 0^+ \\ \hline y' & +ve & 0 & +ve \end{array}$$
 $\therefore$ horizontal point of inflexion at $x = 0$, ie at $(0, 0)$

$$x = 3, \quad \begin{array}{c|c|c|c} x & 3^+ & 3 & 3^- \\ \hline y' & +ve & 0 & -ve \end{array}$$
 $\therefore$ local maximum at $x = 3$, ie at $(3, 27)$

Question 13 (a) (ii)

Criteria	Marks
• Correct solution	2
• Locates the stationary points on the sketch of the curve, or equivalent merit	1

Sample answer:**Question 13 (b) (i)**

Criteria	Marks
• Provides correct solution	2
• Attempts to complete the square, or equivalent merit	1

Sample answer:

$$x^2 - 4x = 12y + 8$$

$$x^2 - 4x + 4 = 12y + 12$$

$$x^2 - 4x + 4 = 12(y + 1)$$

$$(x - 2)^2 = 4(3)(y + 1)$$

$$\therefore \text{focal length} = 3$$

Question 13 (b) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer: $\therefore$ focus (2,2)**Question 13 (c) (i)**

Criteria	Marks
• Provides correct answer	1

Sample answer: $A = 10$ **Question 13 (c) (ii)**

Criteria	Marks
• Provides correct solution	2
• Obtains $5 = 10e^{-163k}$, or equivalent merit	1

Sample answer:

$$M(t) = 10e^{-kt}$$

$$M(163) = 5$$

$$\therefore 5 = 10e^{-163k}$$

$$\frac{1}{2} = e^{-163k}$$

$$-163k = \log_e \left(\frac{1}{2} \right)$$

$$k = \frac{\log_e \left(\frac{1}{2} \right)}{-163}$$

$$\approx 0.004252436$$

Question 13 (d)

Criteria	Marks
• Provides correct solution	3
• Finds the area under the cosine curve, or equivalent merit	2
• Finds the area under $y = x$, or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \text{Area} &= \int_0^1 \sqrt{2} \cos\left(\frac{\pi}{4}x\right) dx - \frac{1}{2} \times 1 \times 1 \\
 &= \sqrt{2} \times \frac{4}{\pi} \left[\sin\left(\frac{\pi}{4}x\right) \right]_0^1 - \frac{1}{2} \\
 &= \frac{4\sqrt{2}}{\pi} \left(\sin \frac{\pi}{4} - \sin 0 \right) - \frac{1}{2} \\
 &= \frac{4\sqrt{2}}{\pi} \left(\frac{1}{\sqrt{2}} - 0 \right) - \frac{1}{2} \\
 &= \frac{4}{\pi} - \frac{1}{2} \text{ square units} \\
 &\text{or } \frac{8 - \pi}{2\pi}
 \end{aligned}$$

Question 14 (a)

Criteria	Marks
• Provides correct solution	3
• Attempts to find a difference in areas or equivalent merit	2
• Applies Simpson's Rule to the existing heights or equivalent merit	1

Sample answer:

The increase in area can be approximated using Simpson's Rule.

$$\begin{aligned}
 \left. \begin{array}{l} \text{Area increase} \\ \text{(Simpson, 5 values)} \end{array} \right\} &= \frac{1}{3} [2 \times (2 - 2) + 2 \times (3 - 2.5) + 8 \times (2.78 - 2.38)] \\
 &= \frac{1}{3} [0 + 1 + 3.2] \\
 &\approx 1.4 \text{ m}^2
 \end{aligned}$$

Question 14 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Obtains a correct expression for A_1 , or equivalent merit	1

Sample answer:

$$A_0 = 100\,000$$

$$A_1 = 0.65 \times 100\,000 + 5000$$

$$A_2 = 0.65 A_1 + 5000$$

$$= 0.65(0.65 \times 100\,000 + 5000) + 5000$$

Question 14 (b) (ii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$A_2 = 0.65^2 \times 100\,000 + 0.65 \times 5000 + 5000$$

$$A_n = 0.65^n \times 100\,000 + 5000(0.65^{n-1} + 0.65^{n-2} + \dots + 1)$$

$$S_n = \frac{1(1 - 0.65^n)}{0.35}$$

$$A_n = 0.65^n \times 100\,000 + 5000 \frac{(1 - 0.65^n)}{0.35}$$

Question 14 (b) (iii)

Criteria	Marks
• Provides correct answer	1

Sample answer:

$$A_{14} = 0.65^{14} \times 100\,000 + 5000 \frac{(1 - 0.65^{14})}{0.35}$$

$$= 14\,491.7\Box$$

$$\approx 14\,500$$

Question 14 (c) (i)

Criteria	Marks
• Provides correct solution	1

Sample answer:

If w is the total width then

$$\text{Area} = 720 = x \times w$$

$$w = \frac{720}{x}$$

$$\text{Perimeter} = 5 \times x + w$$

$$= 5x + \frac{720}{x}$$

Question 14 (c) (ii)

Criteria	Marks
• Provides correct solution	3
• Finds length at stationary point, or equivalent merit	2
• Finds $\frac{d\ell}{dx}$, or equivalent merit	1

Sample answer:

Stationary points occur when

$$\begin{aligned} 0 &= \frac{d\ell}{dx} \\ &= 5 - \frac{720}{x^2} \end{aligned}$$

$$\frac{720}{x^2} = 5$$

$$x^2 = 144$$

$$x = 12 \quad (x \text{ is length so ignore } -12)$$

$$\frac{d^2\ell}{dx^2} = \frac{1440}{x^3}$$

$$\text{at } x = 12 \quad \frac{d^2\ell}{dx^2} = \frac{1440}{12^3} > 0$$

so minimum at $x = 12$

$$\begin{aligned} \ell &= 5 \times 12 + \frac{720}{12} \\ &= 120 \end{aligned}$$

Question 14 (d)

Criteria	Marks
• Provides correct solution	2
• Sums the series, or equivalent merit	1

Sample answer:

$$1 + x + x^2 + x^3 + x^4 = \frac{1 - x^5}{1 - x} \quad a = 1, r = x$$

$$= \frac{x^5 - 1}{x - 1} \quad \text{for } x \neq 1$$

$$\lim_{x \rightarrow 1} \frac{x^5 - 1}{x - 1} = \lim_{x \rightarrow 1} 1 + x + x^2 + x^3 + x^4$$

$$= 5$$

Question 14 (e)

Criteria	Marks
• Provides correct solution	2
• Expresses terms involving powers of 2, or equivalent merit	1

Sample answer:

$$\log 2 + \log 4 + \log 8 + \cdots + \log 512$$

$$= \log 2 + \log 2^2 + \log 2^3 + \cdots + \log 2^9$$

$$= \log 2 + 2\log 2 + 3\log 2 + \cdots + 9\log 2$$

$$= \frac{9}{2}(\log 2 + 9\log 2)$$

$$= \frac{9}{2} \times 10\log 2$$

$$= 45\log 2$$

Question 15 (a)

Criteria	Marks
• Provides correct solution	4
• Evaluates the correct integral for the volume when C_2 is revolved, or equivalent merit	3
• Recognises the sum of two volumes is required and makes progress with both, or equivalent merit	2
• Attempts to find a volume using integration, or equivalent merit	1

Sample answer:

Volume of C_1 rotated:

$$V_1 = \frac{1}{2} \times \frac{4}{3} \pi 2^3$$

$$= \frac{16\pi}{3}$$

Volume of C_2 rotated:

$$V_2 = \pi \int_0^3 y^2 dx$$

$$= \pi \int_0^3 4 \left(1 - \frac{x^2}{9} \right) dx$$

$$= 4\pi \left[x - \frac{x^3}{27} \right]_0^3$$

$$= 4\pi \left(3 - \frac{27}{27} \right)$$

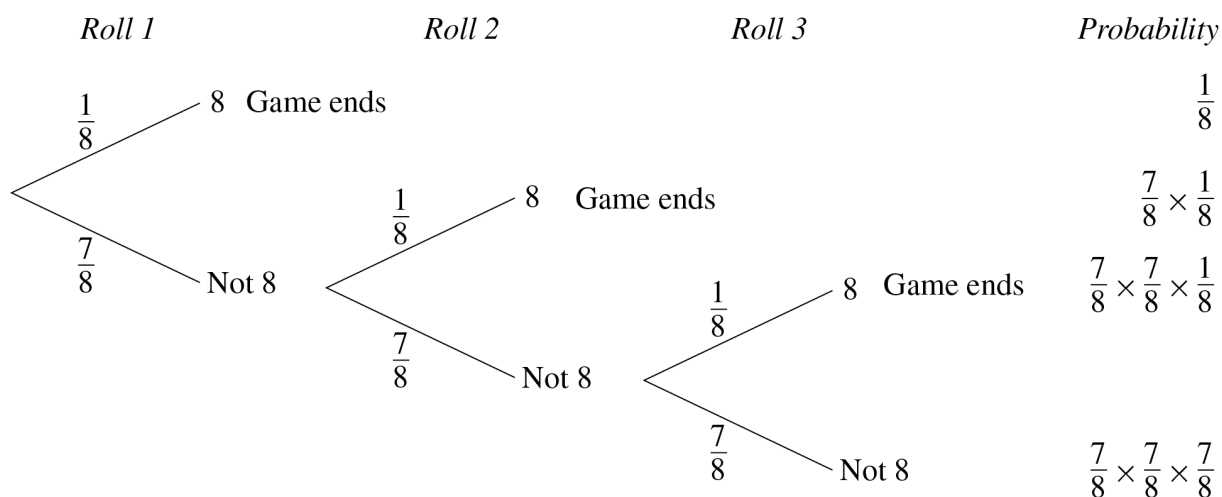
$$= 8\pi$$

$$\therefore V = \frac{16\pi}{3} + 8\pi$$

$$= \frac{40\pi}{3}$$

Question 15 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Provides correct tree diagram, or equivalent merit	1

Sample answer:

$$\therefore P(\text{game ends before 4th roll}) = \frac{1}{8} + \frac{7}{8} \times \frac{1}{8} + \left(\frac{7}{8}\right)^2 \times \frac{1}{8}$$

Alternative solution:

$$P(\text{ends after 1 roll}) = \frac{1}{8}$$

$$P(\text{ends after 2 rolls}) = \frac{7}{8} \times \frac{1}{8}$$

$$P(\text{ends after 3 rolls}) = \left(\frac{7}{8}\right)^2 \times \frac{1}{8}$$

$$\therefore P(\text{ends before 4th roll}) = \frac{1}{8} + \frac{7}{8} \times \frac{1}{8} + \left(\frac{7}{8}\right)^2 \times \frac{1}{8}$$

Question 15 (b) (ii)

Criteria	Marks
• Provides correct solution	3
• Finds an equation or inequality of probabilities that can be solved for n , or equivalent merit	2
• Finds, as a series, the probability that the game ends before the n th roll, or equivalent merit	1

Sample answer:P (ends before n th roll)

$$= \frac{1}{8} + \frac{7}{8} \times \frac{1}{8} + \left(\frac{7}{8}\right)^2 \times \frac{1}{8} + \left(\frac{7}{8}\right)^3 \times \frac{1}{8} + \dots + \left(\frac{7}{8}\right)^{n-2} \times \frac{1}{8}$$

$$= \frac{1}{8} \left[1 + \frac{7}{8} + \left(\frac{7}{8}\right)^2 + \left(\frac{7}{8}\right)^3 + \dots + \left(\frac{7}{8}\right)^{n-2} \right]$$

$$= \frac{1}{8} \times \frac{\left(\left(\frac{7}{8}\right)^{n-1} - 1\right)}{\left(\frac{7}{8}\right) - 1}$$

$$= 1 - \left(\frac{7}{8}\right)^{n-1}$$

$$\text{Let } 1 - \left(\frac{7}{8}\right)^{n-1} = \frac{3}{4}$$

$$\left(\frac{7}{8}\right)^{n-1} = \frac{1}{4}$$

$$(n-1)\log\left(\frac{7}{8}\right) = \log\left(\frac{1}{4}\right)$$

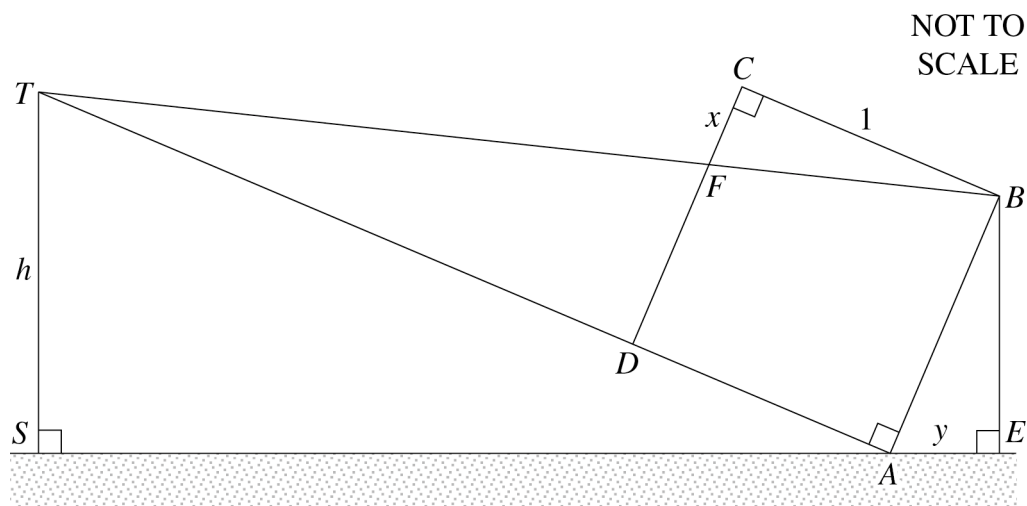
$$n = 1 + \frac{\log \frac{1}{4}}{\log \frac{7}{8}}$$

$$\approx 11.3817\dots$$

$\therefore$ For probability of more than $\frac{3}{4}$, we require $n = 12$

Question 15 (c) (i)

Criteria	Marks
• Provides correct solution	2
• Identifies one pair of equal angles, giving reason(s)	1

Sample answer:In $\triangle s FCB, BAT$ $\angle FCB = \angle BAT$ (both 90° angles in square $ABCD$)Now $AB \parallel DC$ (opposite sides of a square) $\therefore \angle CFB = \angle ABT$ (alternate $\angle s$, $AB \parallel DC$) $\therefore \triangle FCB \parallel \triangle BAT$ (2 pairs of equal $\angle s$).**Question 15 (c) (ii)**

Criteria	Marks
• Provides correct solution	2
• Shows that $\angle TAS$ and $\angle BAE$ are complementary, or equivalent merit	1

Sample answer: $\angle TSA = \angle BAD = \angle AEB = 90^\circ$ $\angle TAS + \angle BAE = 90^\circ$ ($\angle$ sum, straight line SAE) $\angle ABE + \angle BAE = 90^\circ$ ($\angle$ sum $\triangle ABE$) $\therefore \angle TAS = \angle ABE$ In $\triangle s TSA, AEB$ $\angle TAS = \angle ABE$ (above) $\angle TSA = \angle AEB = 90^\circ$ (given) $\therefore \triangle TSA \parallel \triangle AEB$ (2 pairs of equal $\angle s$).

Question 15 (c) (iii)

Criteria	Marks
• Provides correct solution	2
• Obtains $AT = \frac{1}{x}$, or equivalent merit	1

Sample answer:

$$\frac{TS}{AE} = \frac{TA}{AB} \quad (\text{matching sides in similar } \triangle\text{s})$$

$$\frac{h}{y} = \frac{TA}{1}$$

$$\therefore h = y.TA$$

Also $\frac{TA}{BC} = \frac{BA}{FC} \quad (\text{matching sides in similar } \triangle\text{s})$

$$\frac{TA}{1} = \frac{1}{x}$$

$$\therefore TA = \frac{1}{x}$$

Hence $h = y \times \frac{1}{x}$

$$\therefore h = \frac{y}{x}$$

Question 16 (a) (i)

Criteria	Marks
• Provides correct answer	1

Sample answer:When $t = 0$

$$v = 2 - \frac{4}{0+1} = -2$$

Question 16 (a) (ii)

Criteria	Marks
• Provides correct solution	2
• Finds the value of t for which $v = 0$, or equivalent merit	1

Sample answer:

The particle is stationary when $v = 0$.

$$\text{So } v = 0 \Rightarrow 0 = 2 - \frac{4}{t+1}$$

$$\frac{4}{t+1} = 2$$

$$4 = 2(t+1)$$

$$4 = 2t + 2$$

$$2 = 2t$$

$$1 = t$$

So particle is stationary when $t = 1$.

$$\text{acceleration} = \frac{dv}{dt}$$

$$\frac{dv}{dt} = -4(t+1)^{-2} \times -1$$

$$= \frac{4}{(t+1)^2}$$

$$\text{when } t = 1 \quad \frac{dv}{dt} = \frac{4}{(1+1)^2} = \frac{4}{4} = 1$$

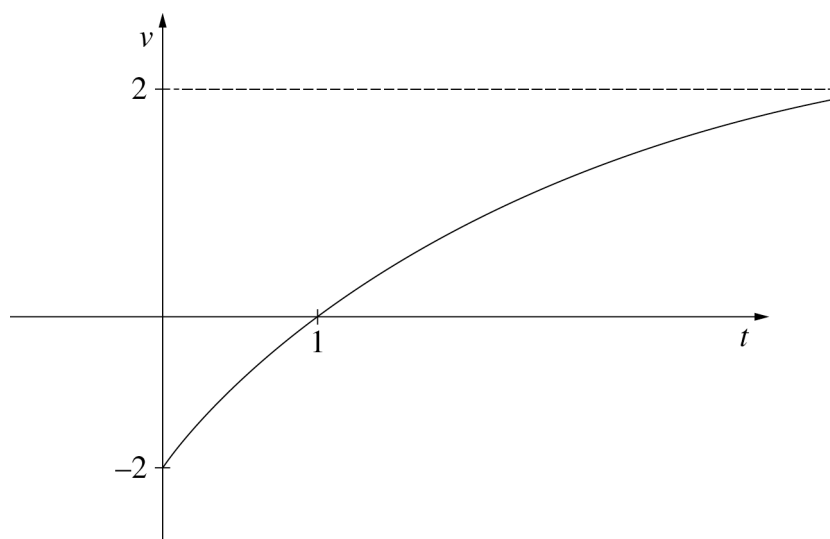
acceleration is 1 ms^{-2} when particle is stationary.

Question 16 (a) (iii)

Criteria	Marks
• Provides correct graph	2
• Describes the behaviour of v for large t , or equivalent merit	1

Sample answer:

As $t \rightarrow \infty, \frac{4}{t+1} \rightarrow 0$
 $\Rightarrow v \rightarrow 2$



Question 16 (a) (iv)

Criteria	Marks
• Provides correct solution	3
• Correctly evaluates $\int_1^7 v dt$, or equivalent merit	2
• Recognises the particle changes direction at $t = 1$, or equivalent merit	1

Sample answer:

Particle changes direction when $t = 1$ travelling in negative direction for $0 \leq t < 1$, so

$$\begin{aligned}
 \text{Distance travelled} &= -\int_0^1 v dt + \int_1^7 v dt \\
 &= -\int_0^1 2 - \frac{4}{t+1} dt + \int_1^7 2 - \frac{4}{t+1} dt \\
 &= -[2t - 4\log(t+1)]_0^1 + [2t - 4\log(t+1)]_1^7 \\
 &= -(2 - 4\log 2 - 0 + 4\log 1) + (14 - 4\log 8 - 2 + 4\log 2) \\
 &= -2 + 4\log 2 + 14 - 4\log 8 - 2 + 4\log 2 \\
 &= 10 + 8\log 2 - 4\log 2^3 \\
 &= (10 + 8\log 2 - 12\log 2) \\
 &= (10 - 4\log 2)
 \end{aligned}$$

Question 16 (b) (i)

Criteria	Marks
• Provides correct solution	2
• Attempts to find $\frac{dy}{dt}$, or equivalent merit	1

Sample answer:

$$y = 200(1 + 19e^{-0.5t})^{-1}$$

$$\text{Rate of growth} = \frac{dy}{dt}$$

$$= -200(1 + 19e^{-0.5t})^{-2} \times \left(\frac{-19e^{-0.5t}}{2} \right)$$

$$= \frac{1900e^{-0.5t}}{(1 + 19e^{-0.5t})^2}$$

Question 16 (b) (ii)

Criteria	Marks
• Provides correct solution with justification	2
• Provides range, or equivalent merit	1

Sample answer:

All terms in $\frac{dy}{dt}$ are positive so y is increasing.

$$\text{when } t = 0 \quad y = \frac{200}{1 + 19} = 10.$$

$$\text{so } y \geq 10 \text{ for } t \geq 0.$$

$$\text{As } t \rightarrow \infty \quad e^{-0.5t} \rightarrow 0$$

$$\text{so } y \rightarrow \frac{200}{1 + 0} = 200$$

$$\text{hence } 10 \leq y < 200.$$

Question 16 (b) (iii)

Criteria	Marks
• Provides correct solution	1

Sample answer:

$$\begin{aligned}
 \frac{y}{400}(200 - y) &= \frac{1}{2(1 + 19e^{-0.5t})} \left(200 - \frac{200}{1 + 19e^{-0.5t}} \right) \\
 &= \frac{100}{(1 + 19e^{-0.5t})} \left(\frac{1 + 19e^{-0.5t} - 1}{1 + 19e^{-0.5t}} \right) \\
 &= \frac{1900e^{-0.5t}}{(1 + 19e^{-0.5t})^2} \\
 &= \frac{dy}{dt}
 \end{aligned}$$

Question 16 (b) (iv)

Criteria	Marks
• Provides correct solution	2
• Recognises the importance of the vertex of the parabola in part (iii), or equivalent merit	1

Sample answer:

$$\begin{aligned}
 \frac{dy}{dt} &= \frac{y}{400}(200 - y) \\
 &= \frac{200y - y^2}{400}
 \end{aligned}$$

which is a quadratic in y with roots at $y = 0$ and $y = 200$. Since the coefficient of y^2 is negative, the quadratic has a max at $y = 100$.

$\therefore$ Population growing fastest when population is $y = 100$.

2016 HSC Mathematics

Mapping Grid

Section I

Question	Marks	Content	Syllabus outcomes
1	1	4.2	P5
2	1	3.1	H5
3	1	9.1, 9.5	P5
4	1	5.1	P4
5	1	12.5, 13.5	H5
6	1	13.3	H5
7	1	13.1	H5
8	1	1.2, 13.2	H5
9	1	1.2, 11.1	H8
10	1	12.2	H3

Section II

Question	Marks	Content	Syllabus outcomes
11 (a)	2	4.3	P5
11 (b)	2	8.9	P7
11 (c)	2	1.2	P3
11 (d)	2	11.1	H5
11 (e)	3	1.4, 6.3, 9.1	P4
11 (f)	2	1.1, 8.4, 13.5	P6, H5
11 (g)	2	13.1, 13.2	H5
12 (a) (i)	2	6.2	P4
12 (a) (ii)	2	6.5	P4
12 (a) (iii)	2	2.3, 6.5	P4
12 (b) (i)	1	2.3	P4
12 (b) (ii)	2	2.4	P4
12 (c)	3	5.5	P4
12 (d) (i)	1	8.8, 12.4	H3, H5
12 (d) (ii)	2	11.1	H3, H5
13 (a) (i)	4	10.2, 10.4	H6
13 (a) (ii)	2	10.5	H6
13 (b) (i)	2	1.3, 9.5	P5
13 (b) (ii)	1	9.5	P5
13 (c) (i)	1	14.2	H4
13 (c) (ii)	2	14.2	H4, H5

Question	Marks	Content	Syllabus outcomes
13 (d)	3	11.4, 13.6	H8
14 (a)	3	11.3	H5, H8
14 (b) (i)	2	7.5	H4, H5
14 (b) (ii)	1	7.5	H4, H5
14 (b) (iii)	1	7.5	H4, H5
14 (c) (i)	1	10.6	H4, H5
14 (c) (ii)	3	10.6	H4, H5
14 (d)	2	7.2, 8.2	P8, H5
14 (e)	2	7.1, 12.2	H3, H5
15 (a)	4	11.4	H8
15 (b) (i)	2	3.3	H5
15 (b) (ii)	3	3.3, 7.2, 12.2	H3, H5
15 (c) (i)	2	2.3, 2.5	H5
15 (c) (ii)	2	2.3, 2.5	H5
15 (c) (iii)	2	2.3, 2.5	H5
16 (a) (i)	1	14.3	H4
16 (a) (ii)	2	14.3	H4, H5
16 (a) (iii)	2	4.2, 10.5	H4, H5, H6
16 (a) (iv)	3	12.5, 14.3	H4, H5
16 (b) (i)	2	12.5, 14.1	H5
16 (b) (ii)	2	4.1, 12.5	H5
16 (b) (iii)	1	1.3, 14.2	H3
16 (b) (iv)	2	9.1, 12.5, 14.2	H5, H7